MILITARY SPECIFICATION

TRANSISTOR, TYPES 2N2631 AND 2N2876

1. SCOPE

1.1 Description.- This specification covers the detail requirements for a silicon, NPN transistor and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Mechanical dimensions and outline.- See figure 1 for transistor type 2N2876 and figure 2 for transistor type 2N2631.

1.3 Absolute maximum ratings.-

V_{CB}	V_{CE}	V_{EB}	I_C Amp		P_C Watts		T_{stg} °C	T_J °C
			2N2631	2N2876	2N2631	2N2876		
80	60	4	1/ 1.5	2/ 2.5	8.75	17.5	-65 to +200	+200

1/ Derate 0.05 W/°C for T_C above 25°C.
2/ Derate 0.1 W/°C for T_C above 25°C.

1.4 Primary electrical characteristics.-

	V_{CE}(sat)		C_{obo}	ft	h_{FE}	P_{out} f = 50 mc		P_{out} f = 150 mc	
	2N2631	2N2876	V_{CB} = 30v		I_C = 300 ma	2N2631	2N2876	2N2631	2N2876
	I_C = 1.5	I_C = 2.5							
Units	Volts	Volts	pf	mc		Watts	Watts	Watts	Watts
Minimum	---	---	---	150	30	7.5	10	3	3
Maximum	1.0	1.0	20	---	150	---	---	---	---

2. APPLICABLE DOCUMENT

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification For

STANDARD

MILITARY

MIL-STD-750 - Tests Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Transistors shall be in accordance with MIL-S-19500 and as specified herein.

FSC 5960

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3.2 Design, construction and physical dimensions.- Transistors shall be of the design, construction and physical dimensions shown on figure 1 for transistor type 2N2876 and figure 2 for transistor type 2N2631.

3.3 Performance characteristics.- Performance characteristics shall be as specified in 4.3 and 4.4

3.4 Marking.- The transistor shall be marked with the "USN" prefix in lieu of the "JAN" prefix. The country of origin may be omitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.3 and 4.4. (Application for qualification tests shall be in accordance with "Provisions Governing Qualification" (see 6.1)).

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in 4.3 and 4.4.

4.3 Group A inspection.- Group A inspection shall consist of the examinations and tests shown in table I. A device having one or more defects shall be counted as one defective.

4.4 Group B inspection.- Group B inspection shall consist of the examinations and tests shown in table II.

4.4.1 Salt atmosphere. The markings shall be legible after the test. There shall be no evidence (when examined with no magnification) of flaking or pitting of the finish or corrosion that will interfere with the mechanical and electrical application of the device.

4.4.2 Destructive tests. Tests listed in subgroups 2, 4 and 6 are considered destructive.

4.4.3 Shock.- The shock waveform shall approximate one half-cycle of a sinewave with a rise time of $0.5\ ^{+0.1}_{-0}$ msec and a decay time of $0.5\ ^{+0.1}_{-0}$ msec. The acceleration of any secondary impacts shall not exceed 20 percent of each primary impact. The amplitude of any response waveform distortion shall not exceed 10 percent of the peak acceleration.

4.5 Input capacitance.- Input capacitance shall be conducted in accordance with Method 3240 of MIL-STD-750 except that the capacitor in the output circuit shall be omitted.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.)

Preparing activity:
Navy - SH
(Project 5960-N488(NAVY))

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Table I - Group A inspection.

Examination or test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 method	Specific Conditions			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071						
Subgroup 2			5				
Breakdown voltage, collector to base	3001 Cond. D	I_C = 100uADC		BV_{CBO}	80	---	Vdc
Breakdown voltage, collector to emitter	3011 Cond. D	I_C = 500mA[1/]		BV_{CEO}	60	---	Vdc
Breakdown voltage, collector to emitter	3011 Cond. A	I_C = 100uA V_{BE} = 1.5V		BV_{CEX}	80	---	Vdc
Latch-up voltage (collector to emitter)		See figure 3		V_{CEXL}	60	---	Vdc
Collector to base cutoff current	3036 Cond. D	V_{CB} = 30Vdc		I_{CBO}	---	0.1	uAdc
Breakdown voltage, emitter to base	3026 Cond. D	I_E = 100uA		BV_{EBO}	4.0	---	Vdc
Collector saturation voltage	3071						
2N2631		I_C = 1.5Adc I_B = 300mAdc		$V_{CE(sat)}$	---	1.0	Vdc
2N2876		I_C = 2.5Adc I_B = 500mAdc		$V_{CE(sat)}$	---	1.0	Vdc
Forward-current transfer ratio	3076	I_C = 300mAdc V_{CE} = 10Vdc PW ≤ 300usec Duty cycle ≤ 2.0%		h_{FE}	30	150	
Subgroup 3			5				
Open circuit output capacitance	3236	V_{CB} = 30Vdc		C_{obo}	---	20	pf
Input capacitance (output open-circuited)	3240 (See 4.5)	V_{EB} = 3Vdc		C_{ibo}	---	300	pf

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Table I - Group A inspection (Cont'd)

Examination or test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 method	Specific Conditions			Min	Max	
Subgroup 3 (Cont'd)							
Extrapolated unity gain frequency	3261	I_C = 250mAdc V_{CE} = 28Vdc f = 100mc		f_t	150	---	mc
Power output		See figure 4 V_{CE} = 28Vdc Pin = 1.0W f = 150mc		P_{out}	3.0	---	Watts
Power output 2N2631		See figure 4 V_{CE} = 28Vdc Pin = 1.0W f = 50mc		P_{out}	7.5	---	Watts
2N2876		V_{CE} = 28Vdc Pin = 2.0W f = 50mc		P_{out}	10.0	---	Watts

1/ Pulse width ≤ 30 usec; duty cycle ≤ 2.0 percent.

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Table II - Group B inspection.

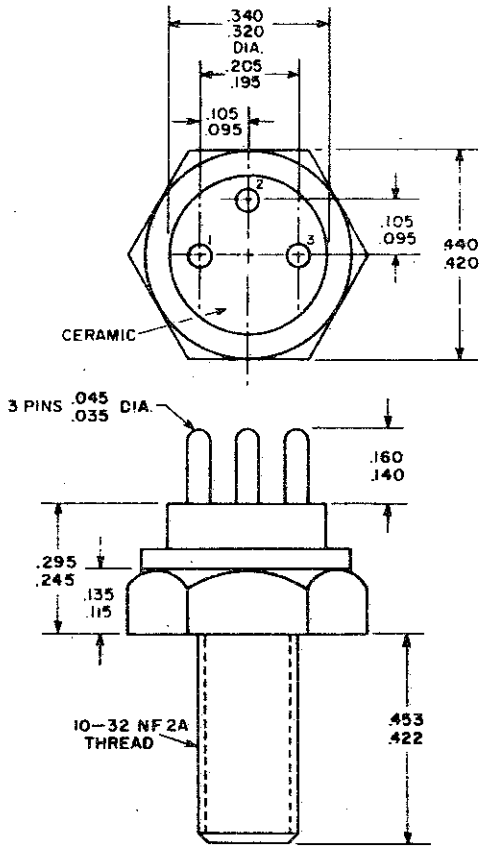
Examination or test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 method	Specific Conditions			Min	Max	
Subgroup 1			20				
Physical dimensions	2066						
Subgroup 2			10				
Solderability	2026						
Thermal shock (temperature cycling)	1051 Cond. C						
Thermal shock (glass strain)	1056 Cond. A						
Moisture resistance	1021						
Subgroup 3			10				
Shock	2016	See 4.4.3 500 G min. 1.0 msec 5 blows each orientation X_1, Y_1, Y_2, Z_1					
Vibration fatigue	2046	Nonoperating					
Vibration, variable frequency	2056						
Constant acceleration	2006	20,000 G					
Subgroup 4			10				
Terminal strength (lead fatigue) 2N2631 only	2036 Cond. E						
Terminal strength (tension) 2N2876 only	2036 Cond. A	Weight = 5 lbs. Time = 10 seconds each terminal					
Terminal strength (stud torque) 2N2876 only	2036 Cond. D2	Torque = 12 in. lbs. Time = 30 seconds					
Subgroup 5			10				
Thermal resistance (junction-to-case) 2N2631	3151			θ_{J-C}	---	20	°C
2N2876				θ_{J-C}	---	10	°C

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Table II - Group B inspection (Cont'd)

Examination or test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 method	Specific Conditions			Min	Max	
Subgroup 6			10				
Salt atmosphere (corrosion)	1041	See 4.4.1					
End points for subgroups 2, 3 and 6:							
Collector to base cutoff current	3036 Cond. D	V_{CB} = 30Vdc		I_{CBO}	---	0.1	uAdc
Power output 2N2631		See figure 4 V_{CE} = 28Vdc Pin = 1.0W f = 50mc		P_{out}	7.5	---	Watts
2N2876		V_{CE} = 28Vdc Pin = 2.0W f = 50mc		P_{out}	10.0	---	Watts
Subgroup 7							
High temperature life (nonoperating)	1031	T_A = 200°C min.	λ = 10				
Subgroup 8							
Steady state operation life 2N2631	1026	T_C = 100°C V_{CE} = 28Vdc I_C = 180mAdc	λ = 10				
2N2876		T_C = 100°C V_{CE} = 28Vdc I_C = 360mAdc					
End points for subgroups 7 and 8:							
Collector to base cutoff current	3036 Cond. D	V_{CB} = 30Vdc		I_{CBO}	---	0.5	uAdc
Power output 2N2631		See figure 4 V_{CE} = 28Vdc Pin = 1.0W f = 50mc		P_{out}	6.8	---	Watts
2N2876		V_{CE} = 28Vdc Pin = 2.0W f = 50mc		P_{out}	9.0	---	Watts

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NOTES:

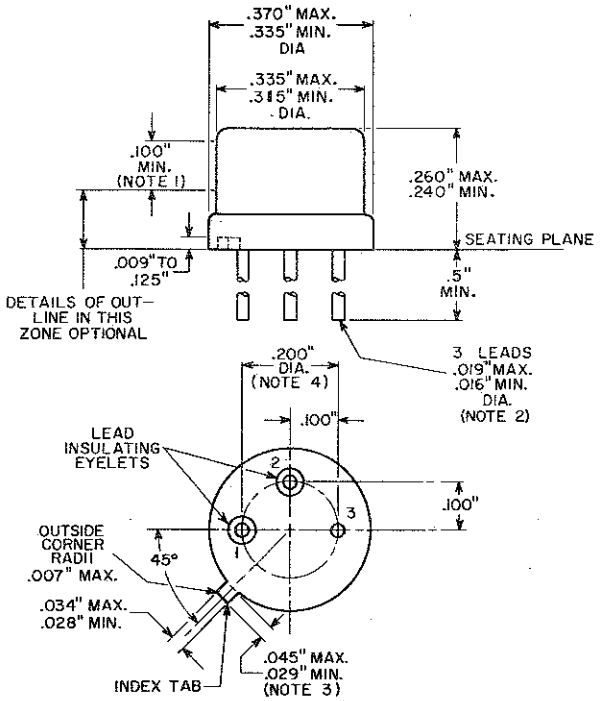
1. The pin spacing permits insertion in any socket having a pin-circle diameter of 0.200 inch and contacts which will accommodate pins having a diameter of 0.035 inch minimum, 0.045 inch maximum.

2. The torque applied to a 10-32 hex nut assembled on the thread during installation should not exceed 12 inch-pounds.

3. This device may be operated in any position.

Figure 1 - Mechanical dimensions and outline of type 2N2876.

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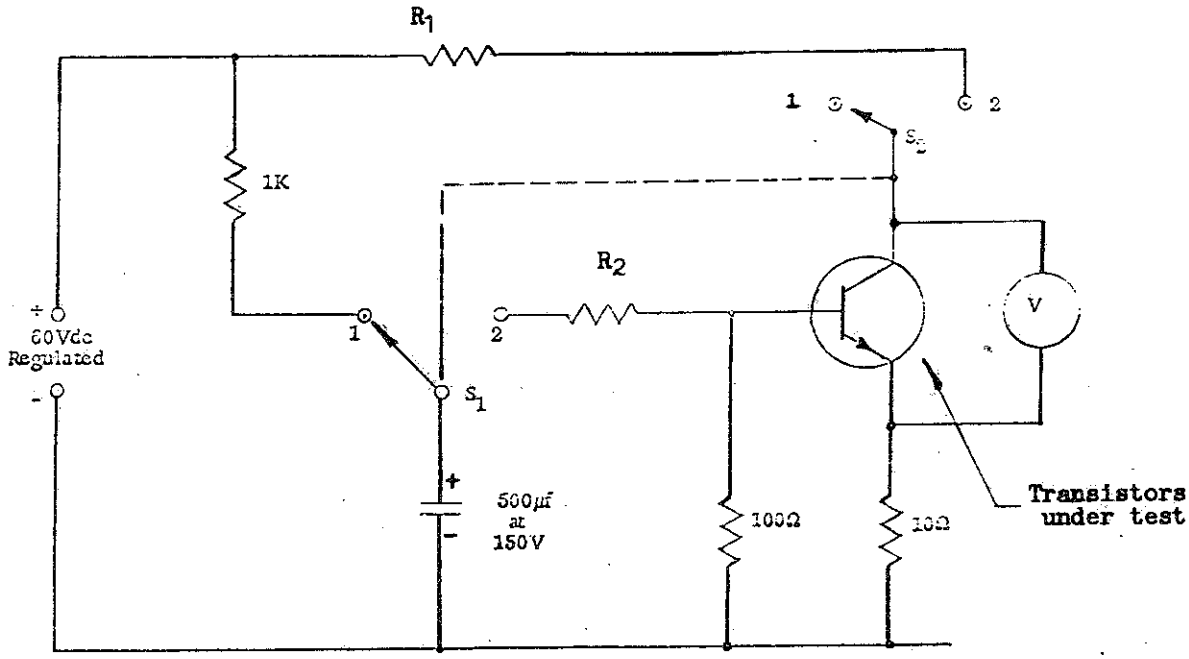


NOTES:

1. This zone is controlled for automatic handling. The variation in actual diameter within the zone shall not exceed 0.010 inch.
2. The specified lead diameter applies in the zone between 0.050 and 0.250 inch from the seating plane. Between 0.250 and 1.5 inches, a maximum of 0.021 diameter is held. Outside of these zones the lead diameter is not controlled.
3. Measured from maximum diameter of the actual device.
4. Leads having maximum diameter (0.019 inch) measured in gauging plane of $0.054\ ^{+0.001}_{-0.000}$ inch below the seating plane of the device shall be within 0.007 inch of their true locations relative to a maximum-width tab.

Figure 2 - Mechanical dimensions and outline of type 2N2631.

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Type	R_1	R_2
2N2631	90 ohms	1000 ohms
2N2876	40 ohms	500 ohms

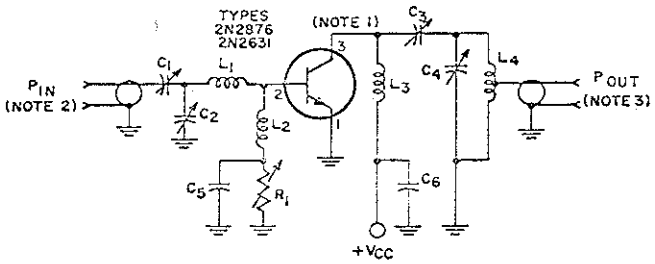
Test method:

1. With switches S_1 and S_2 in position 1, insert transistor into test socket.
2. Move switches S_1 and S_2 to position 2.
   (a) Meter V should rise from zero to 60 volts for good transistors.
   (b) Meter V reading less than 60 volts indicates defective transistor.

Note: S_1 and S_2 = SP - DT Micro-switch.

Figure 3 - Latch-up voltage (collector to emitter) test circuit.

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	For 50mc operation	For 150mc operation
C_1, C_2, C_3, C_4:	8-60 pf	4-40 pf
C_5, C_6:	0.005 uf	0.005 uf
L_1:	8 turns No. 16 wire, 3/8 inch inside diameter by 9/16 inch long	1 turn No. 16 wire, 1/4 inch inside diameter by 3/16 inch long
L_2:	Ferrite choke, Z = 750 ohms ± 20 percent	Ferrite choke, Z = 750 ohms ± 20 percent
L_3:	10 uh	1.5 uh
L_4:	7 turns No. 14 wire, 1/2 inch inside diameter by 7/8 inch long. Tap 2 turns from ground end.	3 turns No. 14 wire, 3/8 inch inside diameter by 3/4 inch long. Tap 1-1/2 turns from ground end.
R_1:	5000 ohms	50 ohms

NOTES:

1. Collector grounded to case in type 2N2631; see terminal diagram.
2. Generator impedance = 50 ohms.
3. Load impedance = 50 ohms.

Figure 4 - Circuit of unneutralized amplifier used to measure power output of types 2N2876 and 2N2631.

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